ChilliFusionNet

Chilli Leaf Disease Classification -- Project Showcase

Generated: 2026-08-25 18:42

Best Validation Accuracy: 0.9982

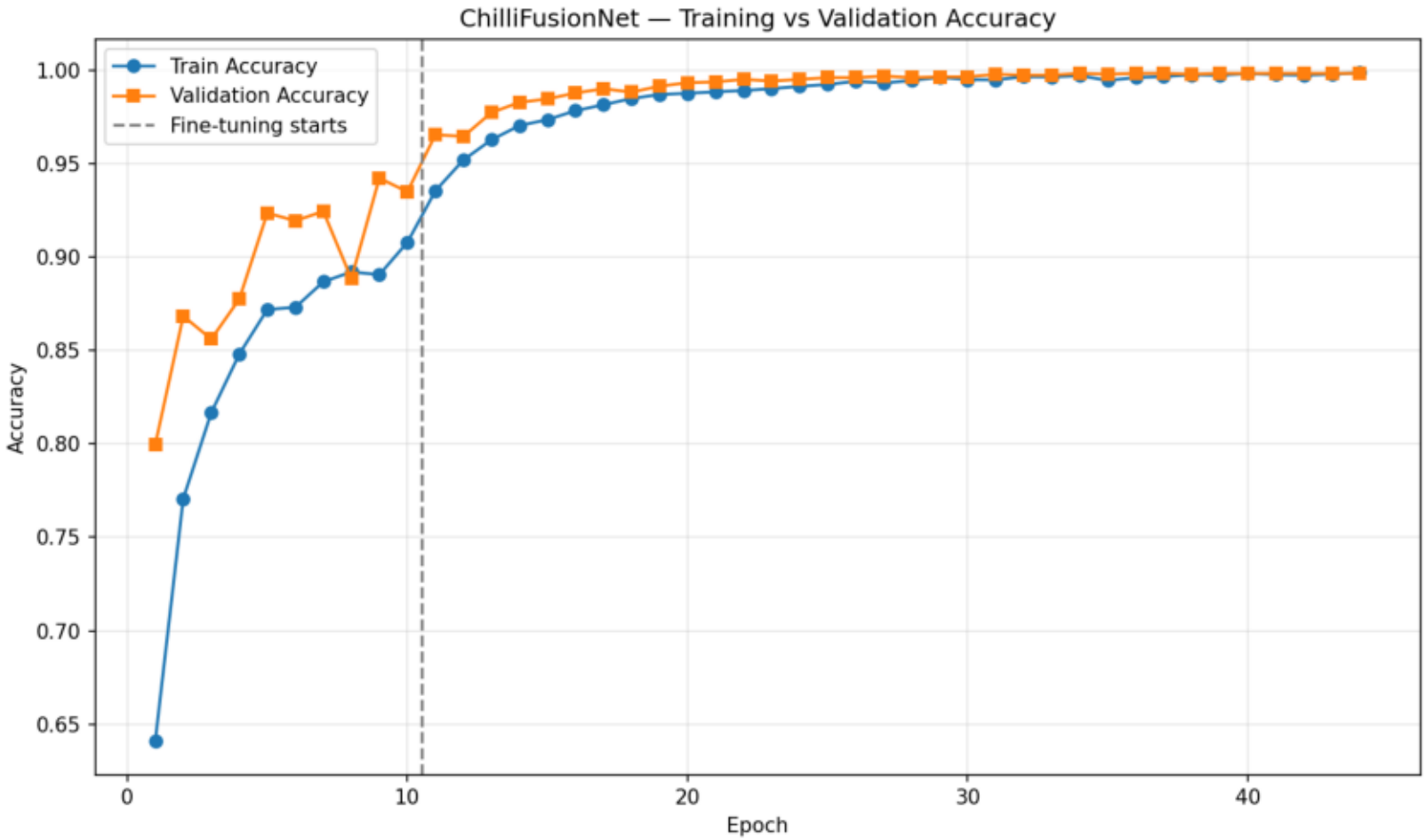
Macro F1: 0.9985

Macro AUC: 1.0

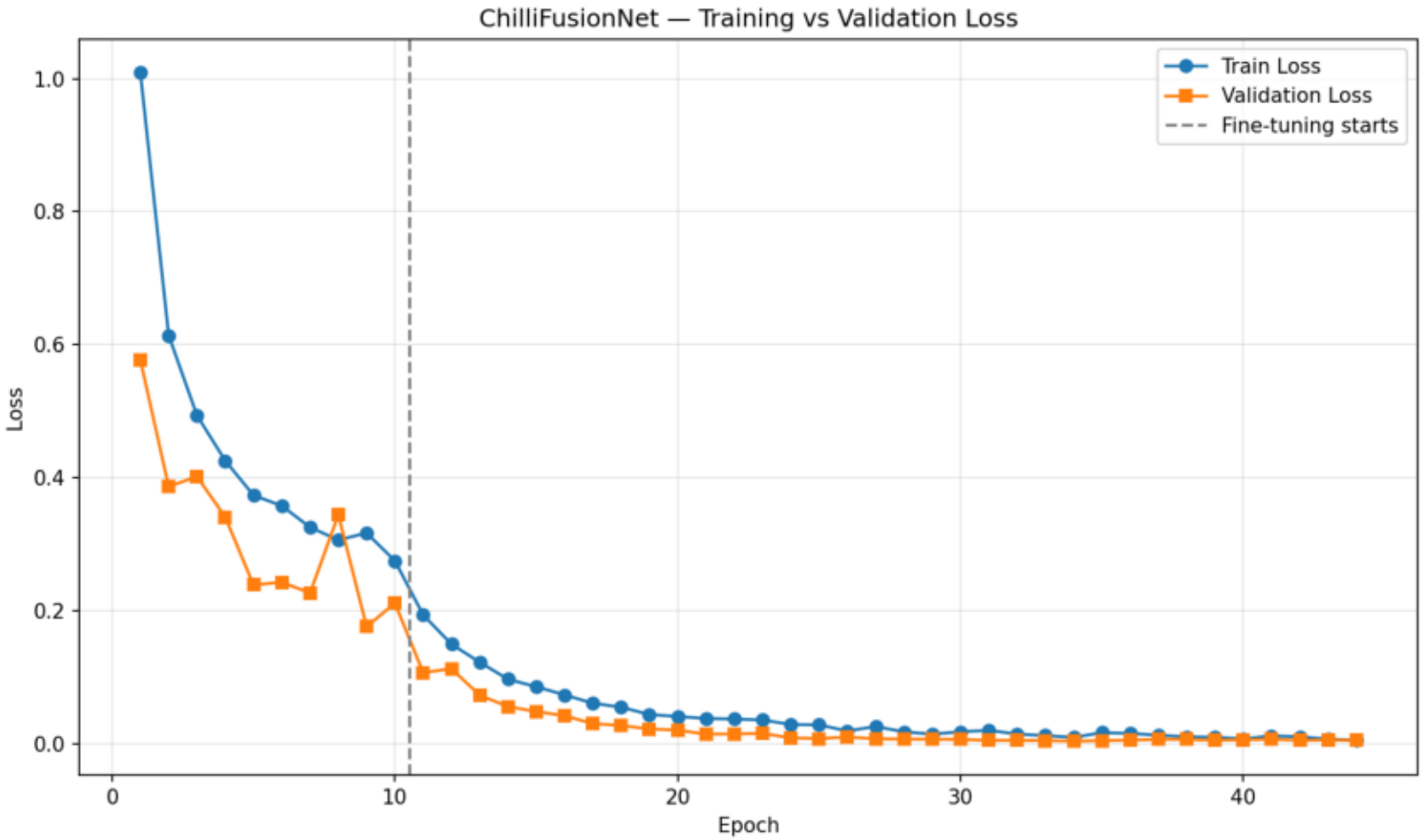
Final Metrics Summary

Metric	Value
Model Name	ChilliFusionNet
Architecture	EfficientNetB4 + CBAM + BiLSTM
Input Size	380x380x3
Number of Classes	5
Initial Epochs (requested)	10
Fine-tuning Epochs (requested)	50
Actual Epochs Run	44
Best Epoch	34
Training Samples	8915
Validation Samples	2195
Test Samples	2225
Best Validation Accuracy	0.9982
Best Validation Loss	0.0039
Final Test Accuracy	0.9987
Test Loss	0.0032
Macro Precision	0.9987
Macro Recall	0.9984
Macro F1	0.9985
Macro AUC	1.0
Macro Sensitivity	0.9984
Macro Specificity	0.9997

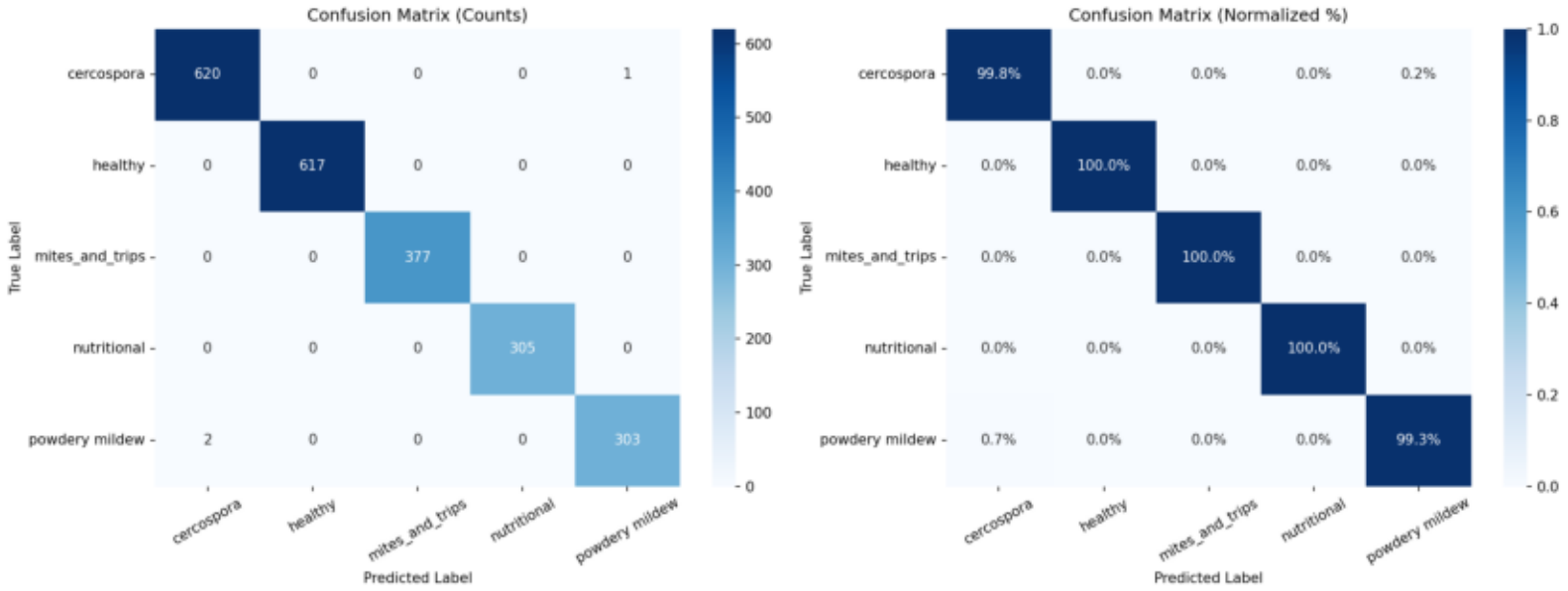
Accuracy Curve



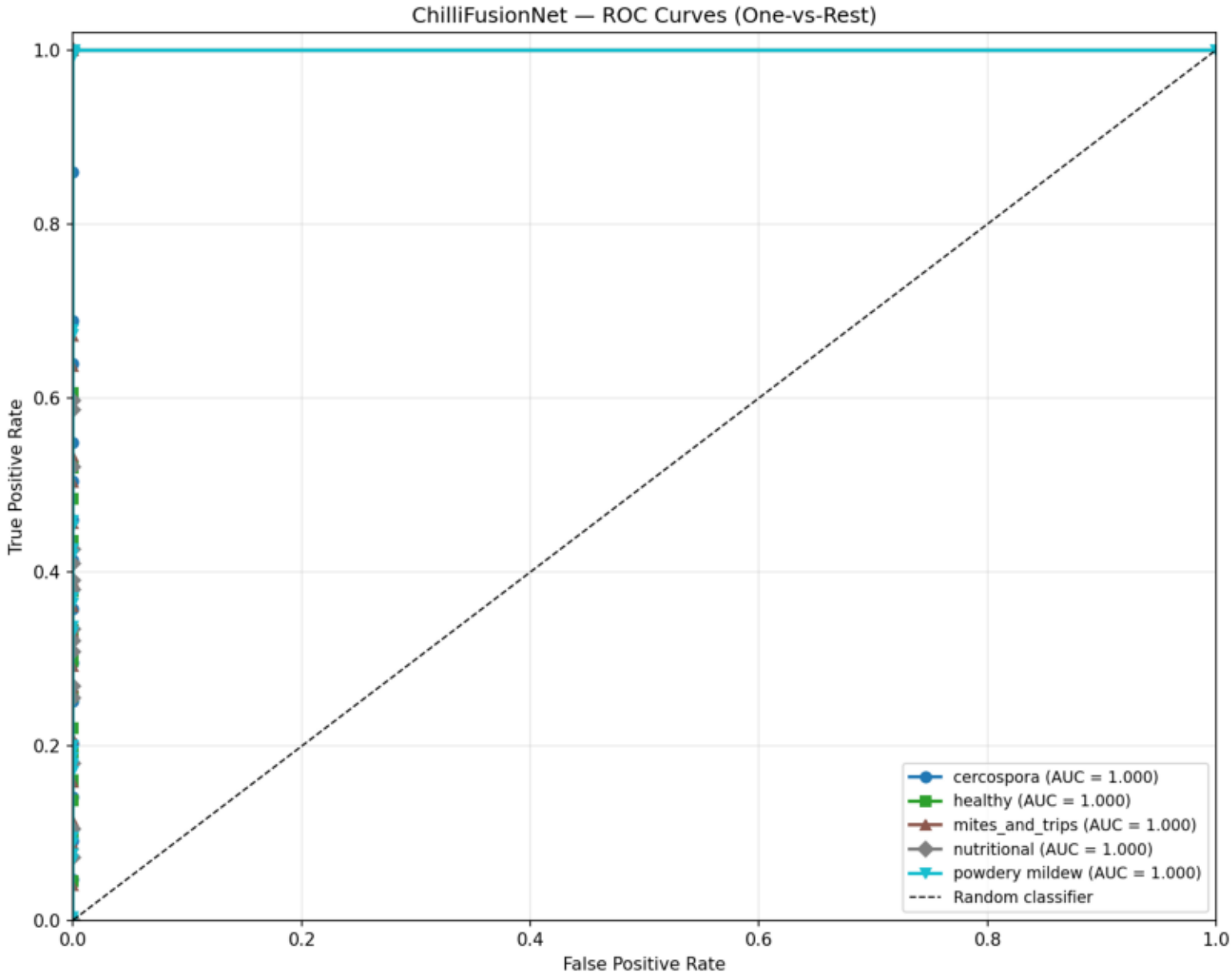
Loss Curve


ChilliFusionNet — Training vs Validation Loss
Train Loss
Validation Loss
Fine-tuning starts
Loss
Epoch
1.0
0.8
0.6
0.4
0.2
0.0
0
10
20
30
40

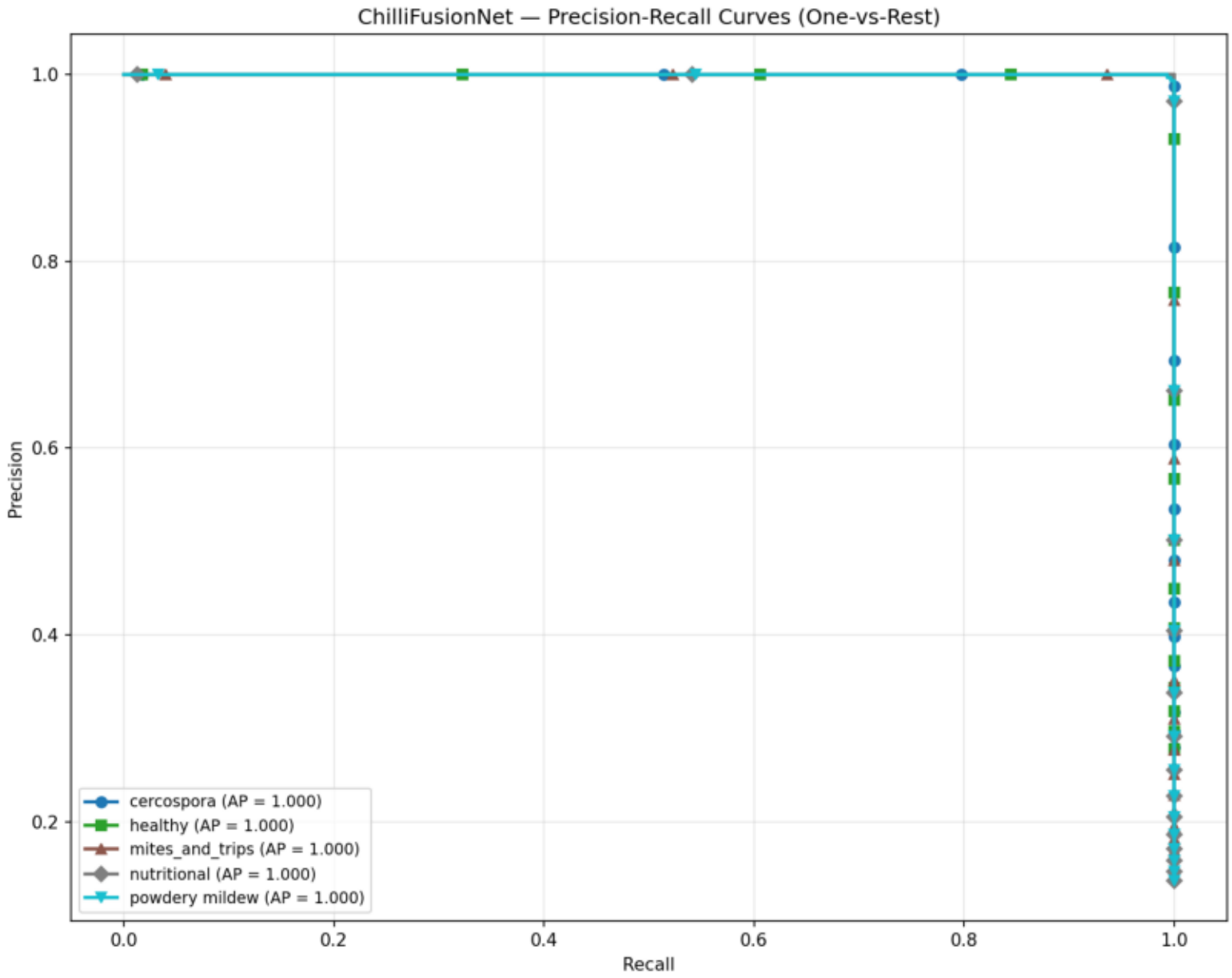
Confusion Matrix



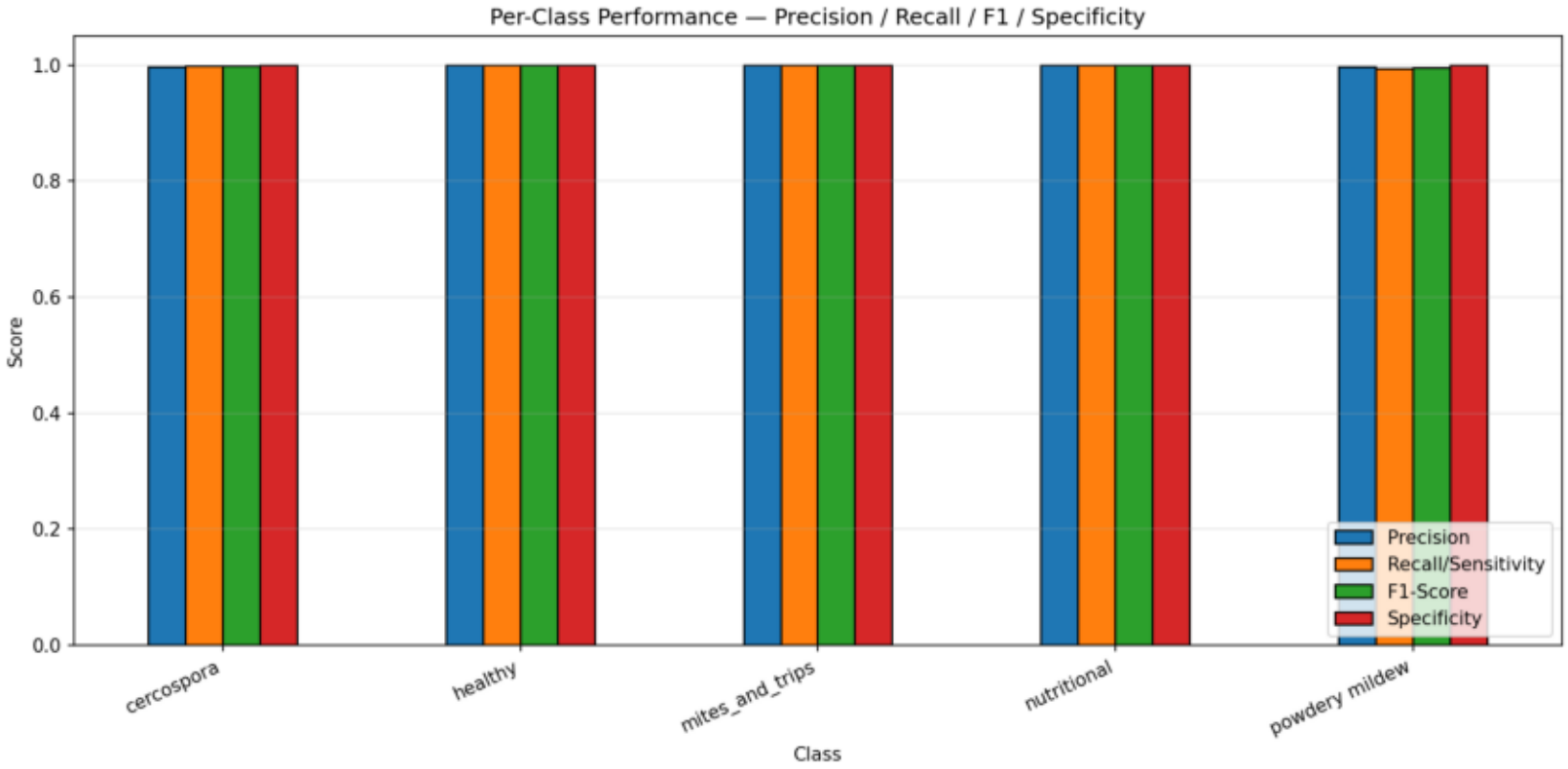
Roc Curve


ChilliFusionNet — ROC Curves (One-vs-Rest)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
cercospora (AUC = 1.000)
healthy (AUC = 1.000)
mites_and_trips (AUC = 1.000)
nutritional (AUC = 1.000)
powdery mildew (AUC = 1.000)
Random classifier

Precision Recall Curve



Per Class Performance



Gradcam Examples

